1 I

1.1 (1)

From the first law:

$$dU = T\,dS + H\,dM.$$

The Gibbs free energy is defined as:

$$G = U - TS - HM.$$

Taking its differential:

$$\begin{aligned} dG &= dU - T\,dS - S\,dT - H\,dM - M\,dH \\ &= -S\,dT - M\,dH. \end{aligned}$$

1.2 (2)

Since G is a state function, its mixed second-order partial derivatives are equal:

$$\frac{\partial^2 G}{\partial H \partial T} = \frac{\partial^2 G}{\partial T \partial H}.$$

Therefore, the Maxwell relation is:

$$\boxed{\left(\frac{\partial S}{\partial H}\right)_T = \left(\frac{\partial M}{\partial T}\right)_H}.$$

1.3 (3)

From the first law:

$$dU = T\,dS + H\,dM.$$

At constant magnetization:

$$\boxed{C_M = \left(\frac{\partial U}{\partial T}\right)_M}.$$

At constant magnetic field:

$$\begin{aligned} \left(\frac{\partial U}{\partial T}\right)_H &= T\left(\frac{\partial S}{\partial T}\right)_H + H\left(\frac{\partial M}{\partial T}\right)_H \\ &= C_H + H\alpha. \end{aligned}$$

Thus:

$$\boxed{C_H = \left(\frac{\partial U}{\partial T}\right)_H - H\alpha}.$$

1.4 (4)

Using the first law and the Maxwell relation:

$$\begin{aligned}\left(\frac{\partial U}{\partial H}\right)_T &= T\left(\frac{\partial S}{\partial H}\right)_T + H\left(\frac{\partial M}{\partial H}\right)_T \\ &= T\alpha + H\chi.\end{aligned}$$

Therefore:

$$\boxed{\left(\frac{\partial U}{\partial H}\right)_T = T\alpha + H\chi}.$$

1.5 (5)

At constant magnetization:

$$dM = \alpha\, dT + \chi\, dH = 0.$$

Therefore:

$$\left(\frac{\partial H}{\partial T}\right)_M = -\frac{\alpha}{\chi}.$$

Using the Maxwell relation:

$$\begin{aligned}C_M &= T\left(\frac{\partial S}{\partial T}\right)_M \\ &= T\left[\left(\frac{\partial S}{\partial T}\right)_H + \left(\frac{\partial S}{\partial H}\right)_T \left(\frac{\partial H}{\partial T}\right)_M\right] \\ &= C_H - \frac{T\alpha^2}{\chi}.\end{aligned}$$

Hence:

$$\boxed{C_M = C_H - \frac{T\alpha^2}{\chi}}.$$

2 II

2.1 (6)

The Hamiltonian is:

$$\mathcal{H} = -\mu h \sum_{i=1}^{N} \sigma_i \epsilon_i.$$

When the magnetic field is uniform, we have $\epsilon_i = 1$ for all i. Therefore:

$$\mathcal{H} = -\mu h \sum_{i=1}^{N} \sigma_i.$$

For a single atom, the two possible energies are:

$$E_+ = -\mu h, \qquad E_- = \mu h.$$

Thus, the single-particle partition function is:

$$\begin{aligned} z_1 &= e^{-\beta E_+} + e^{-\beta E_-} \\ &= e^{\beta\mu h} + e^{-\beta\mu h} \\ &= 2\cosh(\beta\mu h). \end{aligned}$$

Since the atoms are independent:

$$Z = z_1^N = \left[2\cosh(\beta\mu h)\right]^N .$$

The expectation value of the energy is:

$$\begin{aligned} E &= -\frac{\partial \ln Z}{\partial \beta} \\ &= -N\mu h \tanh(\beta\mu h). \end{aligned}$$

2.2 (7)

For a non-uniform magnetic field, the partition function of the i-th atom is:

$$\begin{aligned} \xi_i &= e^{\beta\mu h\epsilon_i} + e^{-\beta\mu h\epsilon_i} \\ &= 2\cosh(\beta\mu h\epsilon_i). \end{aligned}$$

Since the atoms are independent, the total partition function is:

$$Z = \prod_{i=1}^{N} \xi_i.$$

Therefore, the expectation value of the energy is:

$$\begin{aligned} E &= -\frac{\partial \ln Z}{\partial \beta} \\ &= -\sum_{i=1}^{N} \frac{\partial \ln \xi_i}{\partial \beta} \\ &= -\mu h \sum_{i=1}^{N} \epsilon_i \tanh(\beta\mu h\epsilon_i). \end{aligned}$$

2.3 (8)

Using the distribution function $g(\epsilon)$, the sum over all atoms can be replaced by an integral:

$$E = -\mu h \int_0^{\epsilon_{\max}} g(\epsilon)\epsilon \tanh(\beta\mu h\epsilon)\, d\epsilon.$$

The heat capacity is:

$$C = \frac{dE}{dT}.$$

Since $\beta = 1/kT$, we have:

$$\frac{d}{dT}\tanh(\beta\mu h\epsilon) = -\frac{\mu h\epsilon}{kT^2}\frac{1}{\cosh^2(\beta\mu h\epsilon)}.$$

Therefore:

$$C = \frac{(\mu h)^2}{kT^2} \int_0^{\epsilon_{\max}} g(\epsilon) \frac{\epsilon^2}{\cosh^2(\beta \mu h \epsilon)} \, d\epsilon.$$

Substituting $g(\epsilon) = NA\epsilon^\gamma$:

$$\boxed{C = \frac{NA(\mu h)^2}{kT^2} \int_0^{\epsilon_{\max}} \frac{\epsilon^{\gamma+2}}{\cosh^2(\beta \mu h \epsilon)} \, d\epsilon}.$$

2.4 (8)

While $\gamma = 0$ and $\beta \mu \epsilon_{max} \to \infty$:

$$\boxed{C = NA \frac{k^2 T \pi^2}{12 \mu h}}$$